Quiz 3: Applied Natural Language Processing
DATA 441/641

Choose exactly 10 questions to answer! 641 students must answer questions 1,2, and 3, so select 7 additional questions of your choice. 441 students may also choose to answer these questions if they wish.

1. Why might one choose to use a GRU over an LSTM?
   - A) GRUs consistently outperform LSTMs in all sequence modeling tasks.
   - B) GRUs include an additional memory gate compared to LSTMs.
   - C) GRUs are less sensitive to hyperparameter tuning than LSTMs.
   - D) GRUs require fewer parameters and are computationally more efficient.
2. What is the purpose of max pooling in a CNN?
   - A) To selectively emphasize higher-weighted neurons in the convolutional layers.
   - B) To reduce the spatial dimensions of feature maps while retaining important information.
   - C) To remove all non-linearity from feature maps.
   - D) To perform a learnable form of dimensionality reduction.
3. A CNN applies a 3×3 filter with stride 1 on a 32×32 grayscale image without padding, followed by a 5×5 filter with stride 2 and no padding. What will be the final output dimension?
   - A) 28×28
   - B) 16×16
   - C) 13×13
   - D) 10×10
4. What is the primary role of the forget gate in an LSTM cell?
   - A) To determine which parts of the input should be ignored before entering the cell.
   - B) To clear the hidden state at each step for preventing memory overflow.
   - C) It determines how much past information should be erased from the cell state.
   - D) To decide the importance of each feature map within the cell.
5. Why might an RNN or LSTM be used in image processing?
   - A) To efficiently process images as a single-dimensional sequence.
   - B) To process images with a sequential structure, such as handwritten text recognition.
   - C) To detect edges and textures in images.
   - D) To perform data augmentation on image datasets.
6. What is the primary advantage of using convolutional layers instead of fully connected layers for image processing?
   - A) Convolutional layers ensure that each pixel contributes equally to feature learning.

B) They completely eliminate overfitting in deep learning models.
C) They reduce the number of parameters by sharing weights.
D) They replace dropout layers in network architectures.

7. What is the primary issue with standard RNNs when dealing with long sequences?

   A) They require additional attention layers for memory retention.
   B) They can only process sequences up to a fixed length.
   C) They introduce bias due to sequential updates in hidden states.
   D) Vanishing and exploding gradients.

8. Which of the following best describes how RNNs handle sequential data?

   A) They pass information forward without storing any past states.
   B) They rely entirely on convolutional layers to process dependencies.
   C) They maintain a hidden state that carries information from previous time steps.
   D) They encode positional embeddings similar to Transformers.

9. What is a key difference between GRUs and LSTMs?

   A) GRUs introduce an additional cell state to improve memory retention.
   B) GRUs are a purely feedforward mechanism without recurrent components.
   C) GRUs have fewer gates and do not use a separate cell state.
   D) GRUs are strictly designed for NLP tasks and cannot be used for time series.

10. In the context of CNNs, why do deeper networks often perform better?

    A) They replace standard convolutions with self-attention mechanisms.
    B) They eliminate the need for fully connected layers.
    C) They extract more complex hierarchical features.
    D) They prevent overfitting by using a larger number of filters.

11. Which of the following best describes the primary role of convolution in a CNN when applied to text data?

    A) To merge words into a fixed-length representation for fully connected layers.
    B) To identify global semantic patterns in entire text sequences.
    C) To extract local patterns and n-gram features.
    D) To sequentially process each word in the text using recurrence.

12. Which of the following is a major challenge when applying RNNs or LSTMs to images?

    A) They cannot be trained using standard backpropagation algorithms.
    B) They do not perform well on sequential data.
    C) They cannot model non-sequential dependencies.
    D) RNNs are inherently designed for 1D sequential data, making them inefficient for processing high-dimensional images.

13. In an LSTM, why do we use a cell state in addition to a hidden state?

    A) The cell state helps directly compute gradients to avoid vanishing gradients.
    B) The cell state replaces the hidden state in standard RNNs.
    C) The cell state ensures that all past information is stored indefinitely.
    D) To store long-term memory independently from short-term activations.